

Machine Learning Engineer – Panel Interview Instructions

For the panel round of interviews, we would like you to:

1. **Read a bit more about Lore** (see, but do not share, “Resilience and Lore” at <https://careers.lore.co/resilience-lore> – password: lore-careers).
2. **Read a bit about the people** you will be interviewing with (see below).
3. **Select one of the problem** statements that resonates with you and that you would like to discuss. Let me know which one you select!
4. **Prepare a working prototype** to illustrate your approach to solving the chosen problem. You can also use diagrams, flowcharts, bullet points, or any other visual or textual representation that helps illustrate your thought process.
5. **Be prepared to discuss your approach** in detail during every interview including the rationale behind your choices, any potential challenges you foresee, and how you would address them.

Interviewers

You'll be meeting with a selection of three of these folks

Adam Ameele, PsyD is a health service psychologist focused on helping build explorations in Lore to create ongoing resilience-enhancing conversations with users. These dynamic, personalized, and authentic dialogues lead to insights that grow agency and action-oriented mindsets in users.

Alex Ruch, PhD is a machine learning engineer focused on helping us build a flourishing community in Lore. His background in network science and graph machine learning has helped us reimagine how we can unlock connection, reflection, and authentic sharing.

Ben Lansdell, PhD is a machine learning engineer with a background in computational neuroscience. He's been leading the way in the implantation of contextual bandits for our nudge service. Ben's thinking on cybernetic loops has been important in Lore's approach to shifting individual's identities and habits.

Danielle Martin is a writer and prompt engineering expert at Lore. She's widely considered the conscience of our generative AI bot and is relentless in her efforts to get each of us to communicate more clearly.

Denis Sharoukhov, PhD is a physicist by training and a machine learning engineer at Lore. He's been instrumental in instituting novel player rewards mechanisms and is leading the creation of our exploration recommendation engine.

Jessyca Duerr is a storytelling expert at Lore. Recently she's been engaged in ensuring the "why" behind Lore is maximally engaging and clear to our employees, investors, and external partners.

Keesha Erickson, PhD is a machine learning engineer with a background that includes building predictive models in healthcare and spotting fraud in event data. At Lore she's tackling what connection really means and how it can be brought to life to inspire real change.

Victoria Shapiro, J.D. is a regulatory intelligence and strategy expert at Lore. Recently, she's been engaged in shaping policy for us across the fast-moving areas of artificial intelligence, data privacy, and data governance, helping us stay out in front of emerging regulatory trends.

Problem Statements

We're not looking for a production-ready solution. What matters most a clear demonstration of your problem-solving skills including: your reasoning, your trade-offs, and your ability to communicate your thinking clearly. You'll walk through your solution in three interviews. Since we work on cross-functional teams, each interviewer will bring a different lens—some more technical, some less. They'll be looking not just at your machine learning skills, but also at how you operate in an environment that's asynchronous, collaborative, and interdisciplinary.

These scenarios are reflective of challenges we have encountered in the past and you might encounter as a Machine Learning Engineer at Lore. We recognize the scenario cannot be sufficiently detailed in this brief, but we are unable to answer questions prior to the call. However, you are encouraged to make assumptions based on your experience to provide a solution strategy.

Some randomly generated sample data which may help you get started is included. You can access it via

<https://careers.lore.co/mle-case-studies/api/download?token=mle-role-2025> .

It is not guaranteed to be representative but may help as a starting place.

Problem Statement 1: Conversational Evaluation API

Scenario: Imagine you work for a company that leverages conversational data to understand how individuals perceive themselves and how these perceptions evolve over time. These conversations come from one-on-one conversations between a user and a

generative agent (StoryBot). The evolving identity signal drives personalization and recommendation systems.

Task: Develop a REST API endpoint in Python to evaluate the content of conversations and determine what the user believes about themselves or a topic. The endpoint should accept raw conversation data as JSON objects (includes the message and metadata). Consider feature extraction and apply one or more machine learning models. You may use pre-trained models (e.g., from Hugging Face) or build a lightweight model yourself. The output should be in a clear format that can be consumed by different teams to understand the user's evolving beliefs about themselves or different topics (e.g., a team assigning value to conversations, a team developing StoryBot, or a team recommending content in the community). As part of your solution, provide a brief README outlining how to set up and run your API, along with instructions for testing the endpoint.

Problem Statement 2: Real-Time Anomaly Detection in Conversational Data

Scenario: Imagine a platform that processes a continuous stream of conversations from a generative agent (StoryBot). To maintain a safe and engaging environment, you need to monitor these conversations in real time for anomalies such as sudden shifts in sentiment, unusual topic spikes, or atypical user behavior that could indicate emerging issues or opportunities.

Task: Develop a streaming machine learning pipeline in Python that ingests data (message and metadata), does feature extraction and preprocessing in near real time, applies an anomaly detection model (your own or from a 3rd party), and returns alerts when anomalies are detected. For this task, feel free to pick an anomaly you think would be interesting to track (e.g., changes in mood, change in emoji use, shifts in language, use of the phrase “I don’t know”, prompt injection attacks, etc.). As part of your solution, provide a brief README that outlines the architecture, setup instructions, and how to run tests.

Problem Statement 3: Aspirational Recommendation System

Scenario: Ideas and beliefs are shaped in part by the social groups we engage with (hello filter bubbles!). Imagine your goal is to deliver content recommendations that are aligned with who a user wants to become, not just what is entertaining or interesting to them right now. You have access to past conversations and user interaction data.

Task: Build a Python-based system that uses historical conversations and interaction logs to develop a recommendation engine to suggest posts from the online community. Feel free to use existing models or develop your own. Outline what key metrics you think should be monitored, and (if time allows) add some functionality to monitor these to update the

recommendation engine as needed. As part of your solution, provide a brief README that outlines the architecture, setup instructions, and how to run tests.